

AKROMID®

C3 GF 25 1 FR black (7246)

PA66+PA6 GF25 FR(40)

AKROMID® C3 GF 25 1 FR black (7246) is a 25 % glass fiber reinforced, flame retardant and heat stabilised PA6/6.6 blend, colored in black. Listed to UL 94 in black V0 @ 0,4 mm and all-colours V0 @ 0,8mm, the material is based on a flame retardant system that is free of red phosphorus and halogens. Due to its good flowability, an improved surface quality is achieved. This grade is also suitable for applications with high strength requirements in the automotive and E&E markets.

Features

heat stabilised 130 flame retardant surface modified

easy flow E&E

Regulatory



Properties

Modulus

9.100 MPa

Strength

137 MPa

Impact

66 kJ/m²

Mechanical Properties

Tensile modulus ISO 527-2	1 mm/min d.a.m.	9100 MPa
	1 mm/min conditioned	6300 MPa
Tensile stress at break ISO 527-2	5 mm/min d.a.m.	137 MPa
	5 mm/min conditioned	95 MPa
Tensile strain at break ISO 527-2	5 mm/min d.a.m.	3,0 %
	5 mm/min conditioned	5,5 %
Flexural modulus ISO 178	2 mm/min d.a.m.	8800 MPa
	2 mm/min conditioned	6000 MPa
Flexural strength ISO 178	2 mm/min d.a.m.	225 MPa
	2 mm/min conditioned	160 MPa
Flexural strain at break ISO 178	2 mm/min d.a.m.	3,5 %
	2 mm/min conditioned	5 %

Disclaimer

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Charpy impact strength	23°C d.a.m.	66 kJ/m²
ISO 179-1/1eU	23°C conditioned	75 kJ/m²
Charpy notched impact strength	23°C d.a.m.	11 kJ/m²
ISO 179-1/1eA	23°C conditioned	15 kJ/m²

Thermal Properties

RTI electrical	UL 0,4mm Wall thickness	140 °C
UL 746B	UL 0,8mm Wall thickness	150 °C
	UL 1,6mm Wall thickness	150 °C
	UL 3,2mm Wall thickness	150 °C
RTI impact	UL 0,4mm Wall thickness	120 °C
UL 746B	UL 0,8mm Wall thickness	125 °C
	UL 1,6mm Wall thickness	125 °C
	UL 3,2mm Wall thickness	125 °C
RTI strength	UL 0,4mm Wall thickness	135 °C
UL 746B	UL 0,8mm Wall thickness	140 °C
	UL 1,6mm Wall thickness	140 °C
	UL 3,2mm Wall thickness	140 °C
Temperature of deflection under load HDT/A	1,8 MPa	225 °C
ISO 75		
Temperature of deflection under load HDT/B	0,45 MPa	251 °C
ISO 75		
Temperature of deflection under load HDT/C	8 MPa	168 °C
ISO 75		
Melting temperature	DSC, 10K/min	262 °C
ISO 11357-3		
Ball pressure test		> 210 °C
IEC 60695-10-2		
Temperature index for 50% loss of tensile strength	5.000 h	155 °C
IEC 60216	20.000 h	130 °C

Flammability

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Burning rate UL 94	UL 0,4 mm Wall thickness	V-0 Class
	UL 0,8 mm Wall thickness	V-0 Class
	UL 1,6 mm Wall thickness	V-0 Class
	UL 3,2 mm Wall thickness	V-0 Class
Burning rate 5V UL 94	1,6 mm Wall thickness	5VA Class
	3,2 mm Wall thickness	5VA Class
GWFI IEC 60695-2-12	UL 0,4 mm Wall thickness	960 °C
	UL 0,8 mm Wall thickness	960 °C
	UL 1,6 mm Wall thickness	960 °C
	UL 3,2 mm Wall thickness	960 °C
GWIT IEC 60695-2-13	UL 0,4 mm Wall thickness	750 °C
	UL 0,8 mm Wall thickness	750 °C
	UL 1,6 mm Wall thickness	750 °C
	UL 3,2 mm Wall thickness	750 °C
HWI UL 746A	UL 0,4 mm Wall thickness	0 PLC
	UL 0,8 mm Wall thickness	0 PLC
	UL 1,6 mm Wall thickness	0 PLC
	UL 3,2 mm Wall thickness	0 PLC
HAI UL 746A	UL 0,4 mm Wall thickness	0 PLC
	UL 0,8 mm Wall thickness	0 PLC
	UL 1,6 mm Wall thickness	0 PLC
	UL 3,2 mm Wall thickness	0 PLC
Burning rate (<100 mm/min) FMVSS 302	> 1 mm Thickness	+

General properties

Density ISO 1183	23°C	1,37 g/cm³
Humidity absorption ISO 1110	70°C, 62% r.H.	1,8 - 2,0 %
Molding shrinkage ISO 294-4	flow	0,1 - 0,3 %
	transverse	0,5 - 0,7 %

Electrical Properties

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Comparative tracking index IEC 60112	Test liquid A	600 V
Comparative tracking index ASTM D3638	UL	0 PLC
Dielectric strength IEC 60243	UL 3 mm	12 kV/mm
Inclined-Plane Tracking, IPT ASTM D2303-13	UL	1 kV
High voltage arc tracking rate (HVTR) UL 746A	UL	1 PLC
High Volt, Low Current Arc Resistance ASTM D495	UL	6 PLC

Rheological Properties

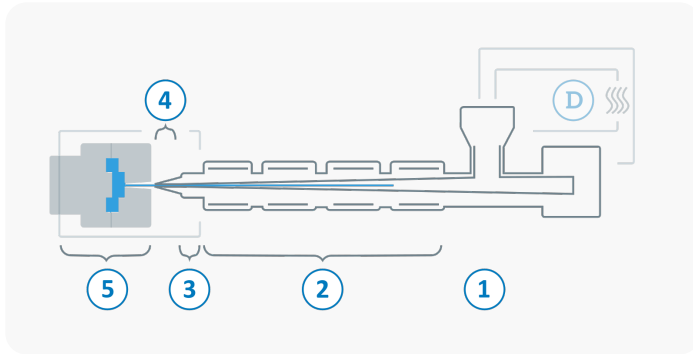
Flowability AKRO	2 mm Thickness	480 mm
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Processing

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only de-humidifying or vacuum dryers. Extensive drying can cause filling problem and surface defects.

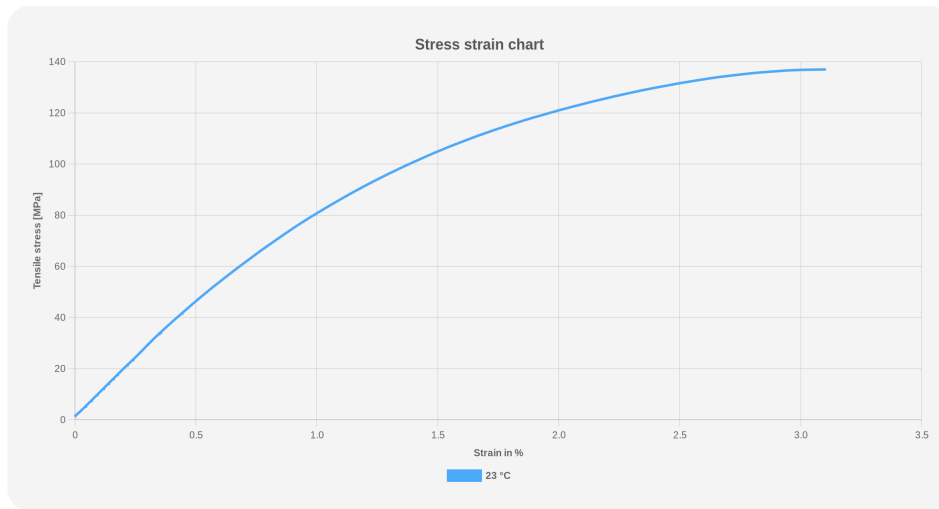


D	Drying time	2 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80 °C
	Processing moisture	0,02 - 0,08 %
1	Feed section	60 - 80 °C
2	Temperature Zone 1 - Zone 4	260 - 290 °C
3	Nozzle temperature	260 - 300 °C
4	Melt temperature	280 - 290 °C
5	Mold temperature	60 - 100 °C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	30 - 100 bar
	Injection speed	medium
	Screw speed	5 - 10 m/min

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Diagrams



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